

Raspberry Pi Zero 2 W + SIM7070 Minimal PPP Setup Guide

1. Enable UART

Edit `/boot/config.txt`:

```
enable_uart=1
dtoverlay=disable-bt
```

Reboot:

```
sudo reboot
```

2. Disable Serial Console

```
sudo raspi-config
```

- Interface Options → Serial Port - Login shell over serial → NO - Enable hardware serial port → YES

3. Wiring / Power

- SIM7070 TX → Pi RX (GPIO15)
 - SIM7070 RX → Pi TX (GPIO14)
 - GND → GND
 - 5V → Pi 5V pin
 - Press or high **PWRKEY** to turn on the modem
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4. Install Required Packages

```
sudo apt update
sudo apt install ppp minicom picocom
```

5. Chat Script `/etc/chatscripts/sim7070.chat`

```
ABORT "BUSY"
ABORT "NO CARRIER"
ABORT "ERROR"
ABORT "NO DIALTONE"
"" AT
OK AT+CPIN?
OK AT+CGDCONT=1,"IP","WSIM"
OK ATD*99#
CONNECT ""
```

Replace `WSIM` with your SIM APN if different.

6. PPP Peer File `/etc/ppp/peers/sim7070`

```
/dev/serial0 115200
connect "/usr/sbin/chat -v -f /etc/chatscripts/sim7070.chat"
noauth
defaultroute
usepeerdns
persist
```

7. Manual Test

1. Kill any previous PPP or chat process:

```
sudo killall -9 pppd chat picocom
```

2. Test modem with picocom:

```
sudo picocom -b 115200 /dev/serial0
```

Type commands:

```
AT
AT+CPIN?
AT+CGDCONT=1,"IP","WSIM"
ATD*99#
```

You should see `OK` responses and eventually `CONNECT`.

8. Start PPP

```
sudo pppd call sim7070 nodetach debug
```

- Watch for:

```
Serial connection established.  
Using interface ppp0  
Connect: ppp0 <--> /dev/serial0  
local IP address ...  
remote IP address ...
```

9. Optional: Python Auto-Start Script

`ppp_connect.py`:

```
import subprocess, time  
  
time.sleep(10)  
subprocess.run(["sudo", "pppd", "call", "sim7070"])
```

- Use systemd to run on boot:

```
sudo nano /etc/systemd/system/ppp-connect.service
```

Content:

```
[Unit]  
Description=PPP connection via SIM7070  
After=network.target  
  
[Service]  
ExecStart=/usr/bin/python3 /home/pi/ppp_connect.py  
Restart=always  
User=pi
```

```
[Install]
WantedBy=multi-user.target
```

Enable:

```
sudo systemctl enable ppp-connect.service
```

10. Optional: Disable Wi-Fi to force cellular

```
sudo systemctl disable wpa_supplicant
sudo systemctl stop wpa_supplicant
```

Notes: - Always ensure `/dev/serial0` is free (no picocom or other process running) before starting PPP. - If `chat` hangs, increase delays using `\d` in the script lines. - Replace `WSIM` with your carrier's APN if needed.